Fiscal Year:	FY 2012	**Task Last Updated:**	FY 01/29/2012
PI Name:	Bouxsein, Mary Ph.D.		
Project Title:	Maintenance of Skeletal Integrity in Spaceflight: Interaction among Partial Weight Bearing, Microgravity and Novel Therapeutics		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	02215-5400	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 Space Biology-FSB-Animal Physiology: NNH09ZTT003N
Start Date:	04/01/2010	**End Date:**	09/01/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
Contact Email:	Sidney.C.Sun@nasa.gov		
Flight Program:	Shuttle/ISS		
Flight Assignment:	STS-135 NOTE: received extension to 9/1/2013 per NSSC (Ed., 4/23/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rutkove, Seward (Beth Israel Deaconess Medical Center)		
Grant/Contract No.:	NNX10AE39G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Despite intensive countermeasures, muscle and bone loss remain among the top medical challenges to extended space missions. Whereas much is known regarding the musculoskeletal responses to microgravity, little is known about the physiological effects of prolonged exposure to partial gravity environments, and whether musculoskeletal deterioration may be mitigated by exposure to partial gravity. To address this knowledge gap, we have developed a novel ground-based model that allows chronic exposure to reduced quadrupedal weightbearing in mice. Using this model will determine whether partial weight bearing attenuates bone and muscle atrophy relative to full hindlimb unloading, and will test the ability of novel spaceflight-compatible therapeutics, one anti-resorptive (antibody to RANKL) and one anabolic (antibody to sclerostin), to inhibit skeletal deterioration across the hypogravity spectrum. Outcome assessments will include in vivo bone density and body composition measurements, along with ex vivo measures of muscle weight, bone microarchitecture, and strength. Serum markers of bone turnover, histomorphometry and gene expression patterns will allow us to delineate the mechanisms underlying the observed skeletal changes. In sum, our proposed studies use both a standardized microgravity analog along with a novel partial gravity model to directly address a key element of the Bioastronautics Roadmap--Accelerated Bone Loss and Fracture Risk. Our ultimate objective is to mitigate the risk for space-induced bone loss and bone fracture associated with missions to low earth orbit, the moon, and eventually Mars. Successful completion of these studies will provide strong rationale for testing these pharmacologic countermeasures in human bed-rest studies and then leading to a trial in astronauts traveling to the International Space Station (ISS). Our findings will also provide insight into the prevention and treatment of skeletal fragility due to immobilization, stroke, cerebral palsy, muscular dystrophy, and spinal cord injury.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The current research has profound impact on Earth-based diseases and disorders resulting from insufficient skeletal loading, such as muscular dystrophy, cerebral palsy, stroke, and other neurodegenerative disorders. Our goal is to characterize the amount of skeletal loading necessary to maintain bone and muscle mass, and to determine the efficacy of novel therapeutics in preventing bone and muscle loss. If successful, this would provide novel insight for the rehabilitation of individuals exposed to musculoskeletal unloading or disuse. Our findings to date indicate that treatment with sclerostin antibody would be likely be effective in preventing disuse induced bone loss in a variety of clinical conditions, such as those listed above.
Task Progress:	To date, we have 1) determined the musculoskeletal effects of partial weight bearing in untreated animals; and 2) completed all testing of the ability of the anti-sclerostin antibody (Scl-AbII) to prevent bone loss. In addition, in a supplementary grant, we have studied the ability of sclerostin antibody to prevent skeletal deterioration during spaceflight by conducting an experiment on the last shuttle mission, STS-135 in July 2011. Additional work on the samples from the spaceflight experiment are pending as is the comparison experiment using the hindlimb unloading model, and thus, we request a no-cost extension of the original one year grant to complete this work. For our ground based work, we studied female wild-type, adult mice (C57Bl/6J, 12 wks of age) subjected to partial weight bearing (20% of body weight, PWB20 and 40% of body weight, PWB40, Lunar and Mars loading analogs), 70% of body weight (PWB70) and hindlimb unloading (HLU) for 21 days. Mice were injected with either sclerostin antibody (Scl-AbII) at 25 mg/kg or vehicle twice weekly. In vehicle treated animals we observed declines in bone and muscle that were largely dose-dependent for the partial weight bearing animals, including declines in bone and muscle even at 70% weight bearing. Scl-AbII completely inhibited the trabecular and cortical bone deterioration due to disuse, with bone properties in HLU and partial weight bearing mice treated with Scl-AbII being at or above values of fully loaded, vehicle treated mice. Histomorphometry and serum bone turnover markers indicate that sclerostin antibody induces new bone formation. As expected, there was no effect of Scl-AbII on muscle loss. For some bone outcomes, the efficacy of Scl-AbII was greater in normally loaded animals than in those exposed to hindlimb unloading or partial weightbearing. Altogether, these early results indicate: 1) partial weight bearing at loading levels up to 70% of body weight is insufficient to prevent bone and muscle loss; and 2) the potent anabolic activity of Scl-AbII, and its efficacy in abrogating disuse-induced bone deterioration. We have not been able to complete Aim 1 due to our inability to obtain huRANKL mice needed to test the antibody to RANKL. The reason for this is that Amgen is unable to provide the mice due to legal issues related to the intellectual property rights surrounding the development of these mice. We have two alternate plans: 1) to test the ability of osteoprotogerin (OPG), a molecule in the RANK-RANKL pathway, to inhibit skeletal deterioration across the spectrum of unloading environments; or 2) to test the ability of a novel anabolic therapy, inhibition of ALK3 receptor signaling. Because of these delays we will request a no—cost extension to the funding period. For our spaceflight experiment (Supplementary grant), we successfully completed a 13-day spaceflight with 30 mice on the shuttle STS-135 and 30 mice maintained as ground controls in the orbital environmental simulator at Kennedy Space Center. In addition, tissues from this experiment were shared with over 60 investigators via NASA’s biospecimen sharing program. Outcome assessments are still ongoing; however, our preliminary data indicate: 1) significant bone and muscle loss in the mice exposed to microgravity, and 2) that a single dose of sclerostin antibody was able to prevent skeletal deterioration and induce new bone formation in microgravity. Further, we found that novel muscle measurements, electrical impedance myography, were sensitive to the muscle atrophy observed following spaceflight. The in vivo portion of the companion ground-based hindlimb unloading (HLU) study will be completed by mid-March, with detailed outcome assessments to follow. Our results to date provide compelling rationale to test sclerostin antibody for its ability to prevent bone loss in human studies of disuse, such as bedrest.
Bibliography Type:	Description: (Last Updated: 04/02/2026)
Abstracts for Journals and Proceedings	Spatz JM, Ellman R, Cloutier AM, Bouxsein ML. "Treatment with anti-sclerostin antibody increases bone mass during hindlimb unloading." Presented at the 18th International Academy of Astronautics Humans in Space Symposium, Houston, TX, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011

Abstracts for Journals and Proceedings	Spatz JM, Ellman R, Cloutier AM, Dwyer D, Stolina M, Ke D, Bouxsein ML. "Treatment with sclerostin antibody increases bone mass and strength during hindlimb unloading." Oral Presentation at 2011 American Society of Bone and Mineral Research Annual Meeting, San Diego, CA, Sept 16-20, 2011. J Bone Miner Res 2011;26 (Suppl 1):S24. Available at: http://www.abstracts2view.com/asbmr/view.php?nu=ASBMR11L_A11006326-65 ; accessed 1/30/2012. , Sep-2011
Abstracts for Journals and Proceedings	Ellman R, Spatz J, Cloutier A, Bouxsein ML. "Bone and muscle loss following partial weightbearing is linearly related to loading level." Presented at the 2011 American Society of Bone and Mineral Research Annual Meeting, San Diego, CA, Sept 16-20, 2011. J Bone Min Res 2011;26(Suppl 1):S265. Available at: http://www.abstracts2view.com/asbmr/view.php?nu=ASBMR11L_A11006364-89 ; accessed 1/30/2012. , Sep-2011
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Abstracts for Journals and Proceedings	Ellman R, Ferguson VL, Livingston EW, Lemus M, Louis LD, Spatz JM, Stodieck LS, Bouxsein ML, Bateman TA. "Effects of spaceflight on bone mineral density and microarchitecture in mice: results from STS-135." Presented as an Oral Presentation at the 2012 NASA Human Research Program Investigators' Workshop, Houston, TX, Feb 14-16, 2012. 2012 NASA Human Research Program Investigators' Workshop, Houston, TX, Feb 14-16, 2012. , Feb-2012
Abstracts for Journals and Proceedings	Sung M, Spieker A, Li J, Ellman R, Spatz J, Bouxsein ML, Rutkove SB. "Electrical impedance alterations in muscle from mice on board STS-135." Oral Presentation at the 2012 NASA Human Research Program Investigators' Workshop, Houston, TX, Feb 14-16, 2012. 2012 NASA Human Research Program Investigators' Workshop, Houston, TX, Feb 14-16, 2012. , Feb-2012